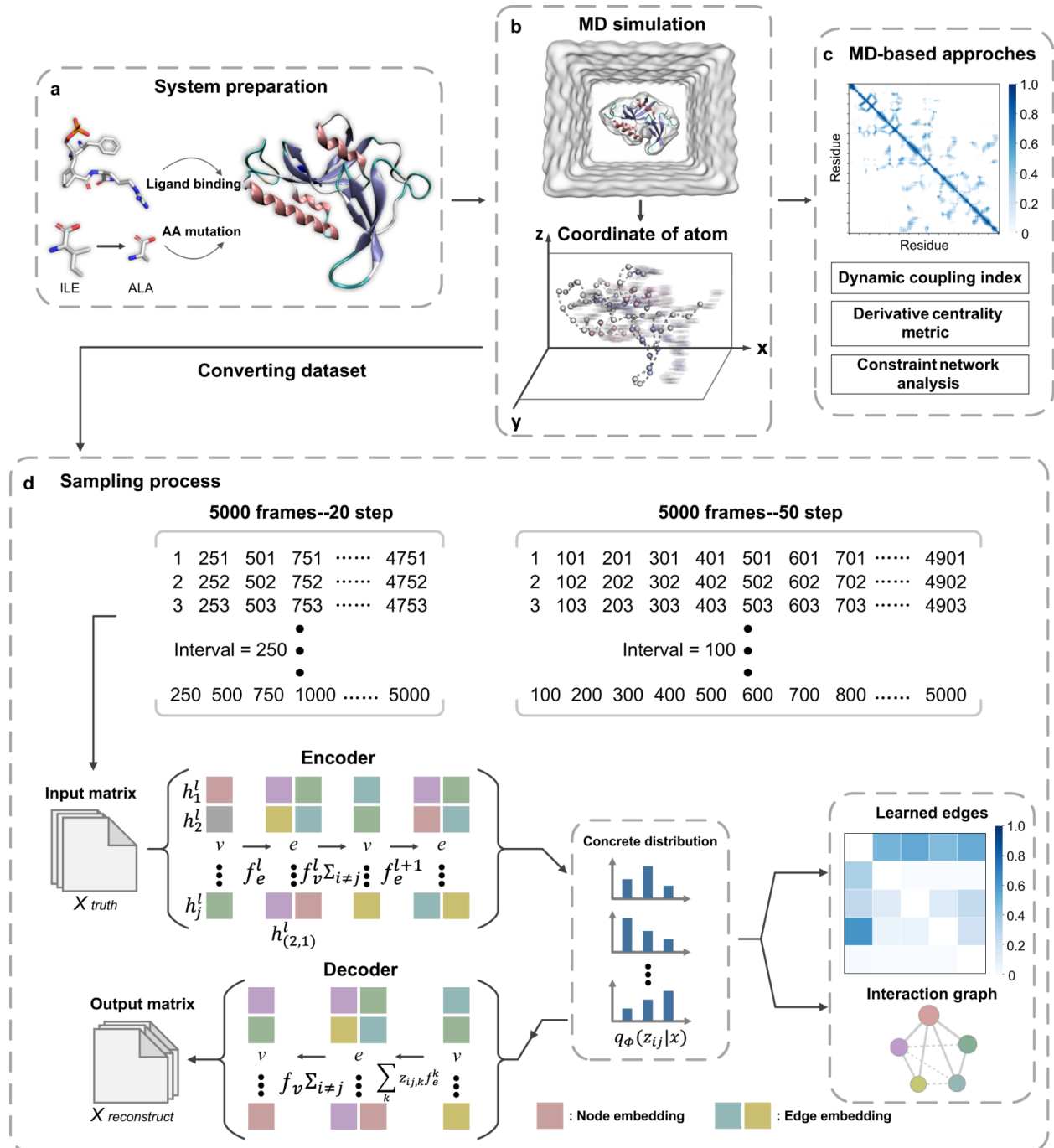


Jingxuan Zhu, Juexin Wang, Weiwei Han, and Dong Xu Neural relational inference to learn long-range allosteric interactions in proteins from molecular dynamics simulations. *Nature Communications*, 13:1661, 2022.

Protein allostery is a biological process facilitated by spatially long-range intra-protein communication, whereby ligand binding or amino acid change at a distant site affects the active site remotely. Molecular dynamics (MD) simulation provides a powerful computational approach to probe the allosteric effect. However, current MD simulations cannot reach the time scales of whole allosteric processes. The advent of deep learning made it possible to evaluate both spatially short and long-range communications for understanding allostery. For this purpose, we applied a neural relational inference model based on a graph neural network, which adopts an encoder-decoder architecture to simultaneously infer latent interactions for probing protein allosteric processes as dynamic networks of interacting residues. From the MD trajectories, this model successfully learned the long-range interactions and pathways that can mediate the allosteric communications between distant sites in the Pin1, SOD1, and MEK1 systems. Furthermore, the model can discover allostery-related interactions earlier in the MD simulation trajectories and predict relative free energy changes upon mutations more accurately than other methods.



The process of inferring an interaction graph by reconstructing an MD simulation trajectory. The process includes the system preparation of a ligand-binding complex or mutant protein structure with allostery (a), the MD simulation of a prepared allosteric system to obtain the trajectory with the dynamic 3D coordinates (b), the conventional analysis for the trajectory (c), and the sampling and training using the NRI model with two jointly trained components (d).